

SELF-HEALING ENTERPRISE AUTOMATION MESH

Autonomous AI for Enterprise DevOps – Zero-Downtime Infrastructure & Self-Healing Pipelines

Team: HealMesh

Presented By: Anwesha Mohapatra, Shreya Gupta, Kasvi Gupta and Elina Somnath

Introduction

The Enterprise DevOps Crisis:

Modern enterprises run on complex microservices, CI/CD pipelines, and data systems that fail constantly — from dependency conflicts to configuration drift. Every minute of downtime costs thousands, yet fixes take hours.

Why Current Tools Fall Short:

Traditional monitoring alerts you after failure, but diagnosis and remediation still require manual intervention by scarce SRE talent. Cloud AIOps tools demand internet connectivity and compromise data privacy.

Our Vision: Autonomous Self-Healing:

What if infrastructure could heal itself — detecting failures, diagnosing root causes, and redeploying fixes autonomously, all while keeping sensitive data on-premise? Built on OpenClaw's local-first AI agents, this is no longer science fiction.



WHY IT'S CRITICAL



Engineering Hours Wasted on Firefighting

68% of SRE/DevOps time spent on manual triage and incident response. Alert fatigue causes critical warnings to be missed in noisy environments. Autonomous agents handle routine failures, freeing engineers for strategic work.



Silent Data Pipeline Degradation

Schema drift in ETL pipelines corrupts analytics dashboards silently. No autonomous guardians exist to detect and patch data config mismatches. Downstream business decisions made on corrupted data cost millions.



Manual Remediation is Too Slow

Human response time: hours to days for complex dependency conflicts. Kubernetes failures require deep expertise not available 24/7. AI agents diagnose and fix in minutes, iterating until stable deployment.

RESEARCH GAP

No Air-Gapped Autonomous Diagnosis:

Current AIOps tools (Datadog, Dynatrace) require cloud connectivity for AI processing. Enterprise security policies forbid sending logs/credentials outside VPC.

Gap: Zero local-first, privacy-compliant autonomous remediation platforms exist.

No Persistent Learning Across Failures:

Traditional runbooks and ChatOps reset after each incident — no institutional memory. Engineers repeat the same debugging steps because knowledge isn't retained.

Gap: No system that learns from every outage to prevent future repeats.



No Unified DevOps + Data Self-Healing:

CI/CD tools, Kubernetes operators, and ETL monitors work in silos. A failure in one layer isn't correlated with root causes in another.

Gap: No single agentic layer that heals pipelines, clusters, AND data schemas together.

Objectives



Eliminate Configuration Drift

1. 60% of outages stem from YAML misconfigs and dependency conflicts.
2. Our agents auto-detect and patch before failures cascade to production.
3. Target: 80% reduction in configuration-related incidents.



Slash Mean Time to Recovery

1. Current industry average: 4+ hours for critical outages.
2. Autonomous diagnosis + remediation cuts this to under 90 minutes.
3. Persistent memory prevents repeating bad patches across cycles.



Ensure Business Continuity

1. Enterprise downtime costs \$300K+/hour in lost revenue and reputation.
2. Self-healing mesh maintains 99.99% uptime without human intervention.
3. Engineering teams shift from firefighting to innovation and architecture.

TYPES OF AGENTS



CI/CD Sentinel Agent

Monitors n8n pipeline triggers, parses dependency trees on build failures, flags exact version conflicts in requirements.txt or package.json files. Auto-generates YAML patches and validates against K8s schema before deploy.



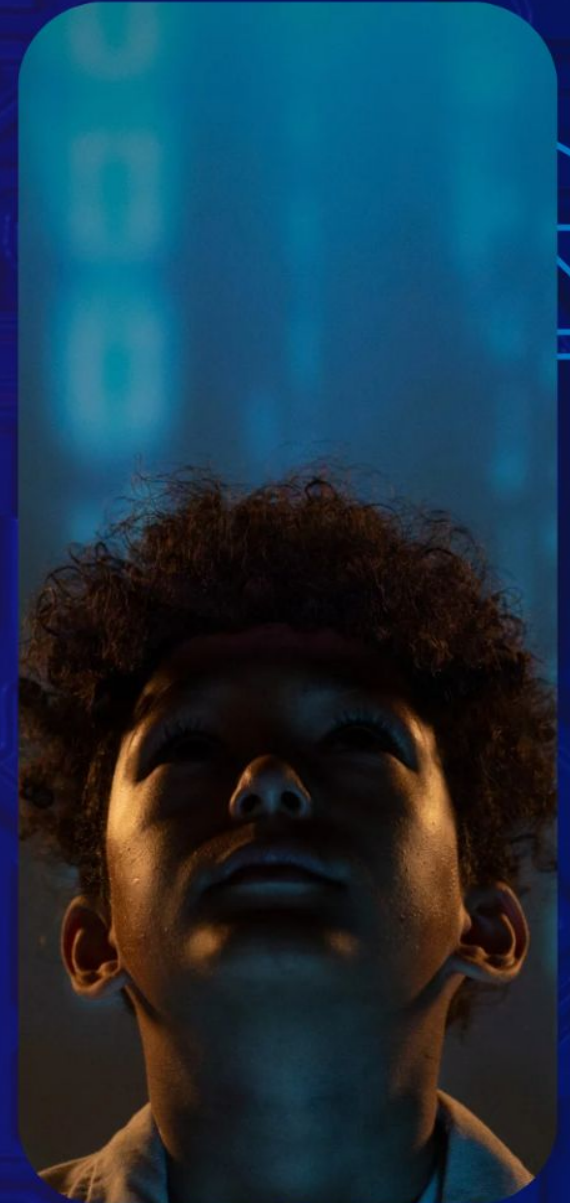
Kubernetes Remediation Agent

Rewrites broken YAML/Helm charts, patches Dockerfiles autonomously, executes safe rollouts via Kubernetes API with policy validation gates. Iterates through fixes until stable deployment achieved without downtime.



Schema Drift Steward Agent

Monitors data pipelines for column/type mismatches in real-time, auto-generates dbt/Airflow config patches before dashboards break. Alerts data engineers with proposed fixes via Slack/Teams integration.



OPENCLAW INTEGRATION



Local-First Diagnosis (Air-Gapped)

Runs entirely on-prem or in VPC — no sensitive logs leave your environment. Hosts Ollama models locally for privacy-compliant failure analysis. Zero egress cost for AI layer, SOC2/GDPR ready out of the box.



Persistent Memory & Learning

Remembers every failure, patch attempt, and engineer override across sessions. Refines heuristics over time to avoid repeating ineffective fixes. Builds cluster-specific knowledge graph for faster diagnosis.



MCP Skill Ecosystem

Plug-in agents for Kubernetes, Helm, Terraform, dbt, Airflow, GitLab. Browser skill scrapes documentation, GitHub issues, and Stack Overflow. Terminal skill executes kubectl, helm, git commands with safety guards.



KEY FEATURES



Real-Time Failure Detection & Alerting

n8n webhooks monitor K8s events, container logs, CI/CD pipelines, ETL jobs. Multi-channel alerts via Slack, Discord, Teams, email with severity levels. Dashboard shows workflow health, redeploy attempts, and MTTR metrics.



AI Root-Cause Diagnosis & Remediation

OpenClaw agents analyze logs, dependency graphs, configs using local/cloud LLMs. Rewrites YAML, adjusts Helm charts, patches Dockerfiles autonomously. Policy validation ensures fixes comply with quotas, RBAC, security baselines.



Continuous Learning & Improvement Loop

Every remediation logged with success/failure outcome and engineer feedback. Agent heuristics improve over time, redundant failures drop 70% annually. Persistent memory prevents repeating bad patches across clusters and teams.

CORE TECHNOLOGIES



n8n Workflow Orchestration

Self-hosted automation platform with 350+ integrations for enterprise tools. Monitors Kubernetes events, CI/CD pipelines, and container logs via webhooks.

Triggers OpenClaw agents on failures, executes redeployments via K8s API.



OpenClaw Autonomous Agent Runtime

Local-first AI runtime with persistent memory across failure cycles. MCP (Model Context Protocol) skill ecosystem for K8s, Helm, Terraform, dbt.

Connects LLMs (Ollama/cloud) to real tools: APIs, terminals, Git repos, Slack.



Kubernetes Self-Healing Engine

Container orchestration with RBAC, namespaces, and resource quotas.

Prometheus/Grafana metrics for real-time cluster health monitoring.

ArgoCD GitOps integration for declarative, auditable configuration changes.

DEMO



1. Screen/Flow

A cinematic, dark-themed Mesh Control Center seamlessly flows between real-time analytics, autonomous healing visualizations, and live system monitoring—creating a cohesive, professional SaaS experience powered by interconnected AI modules.



How User Interacts

Users centrally control the entire mesh via a left-side operational menu, interacting with live agent chat channels and automated incident updates to observe AI-generated fixes, monitor health, and approve remediations from a single command dashboard.



3. Key Moments

The "Self-Healing Mesh Flow" visualization showcases AI agents autonomously diagnosing and repairing Kubernetes failures in real time, while live communication feeds display automated rollbacks and recovery actions—culminating in strong metrics that prove reliability and futuristic operational capability.

POLICY & SECURITY



RBAC & Access Control Enforcement

Validates all AI-proposed changes against Kubernetes RBAC policies. Prevents unauthorized namespace access, privilege escalation, secret exposure. Audit trail logs every agent action with user attribution and timestamp.



Resource Quota & SLO Validation

Blocks deployments exceeding CPU/memory limits or cluster capacity. Checks against Service Level Objectives (SLOs) before executing changes. Auto-scales resources or suggests optimizations if quotas insufficient.



Manual Override & Approval Workflows

Critical changes require human approval via Slack/Teams interactive buttons. GitOps integration creates pull requests for infrastructure-as-code changes. Engineers can reject, modify, or approve AI fixes with full context.

Platform Teams & DevOps Engineers

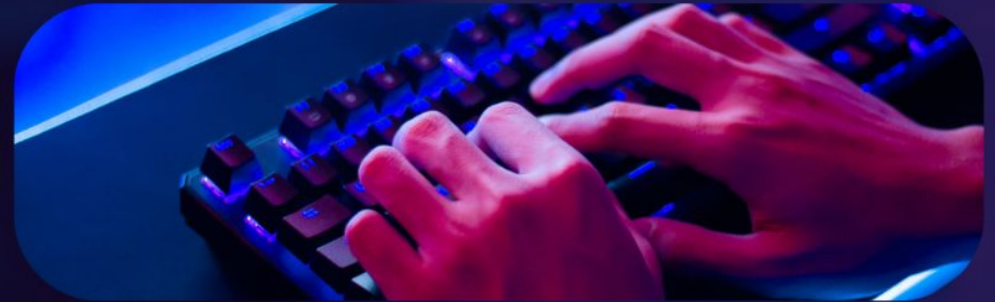
Zero-config self-healing handles 80% of routine outages autonomously. AI co-pilot triages logs, proposes fixes, executes rollouts — focus shifts to architecture. Morning digest highlights cluster health, predicted hotspots, and recommendations.

Data Engineers & Analytics Teams

Schema drift auto-detected before corrupted data breaks downstream dashboards. ETL configs patched automatically with dbt/Airflow updates and SQL migrations. Downstream impact analysis prevents business decision errors from bad data.

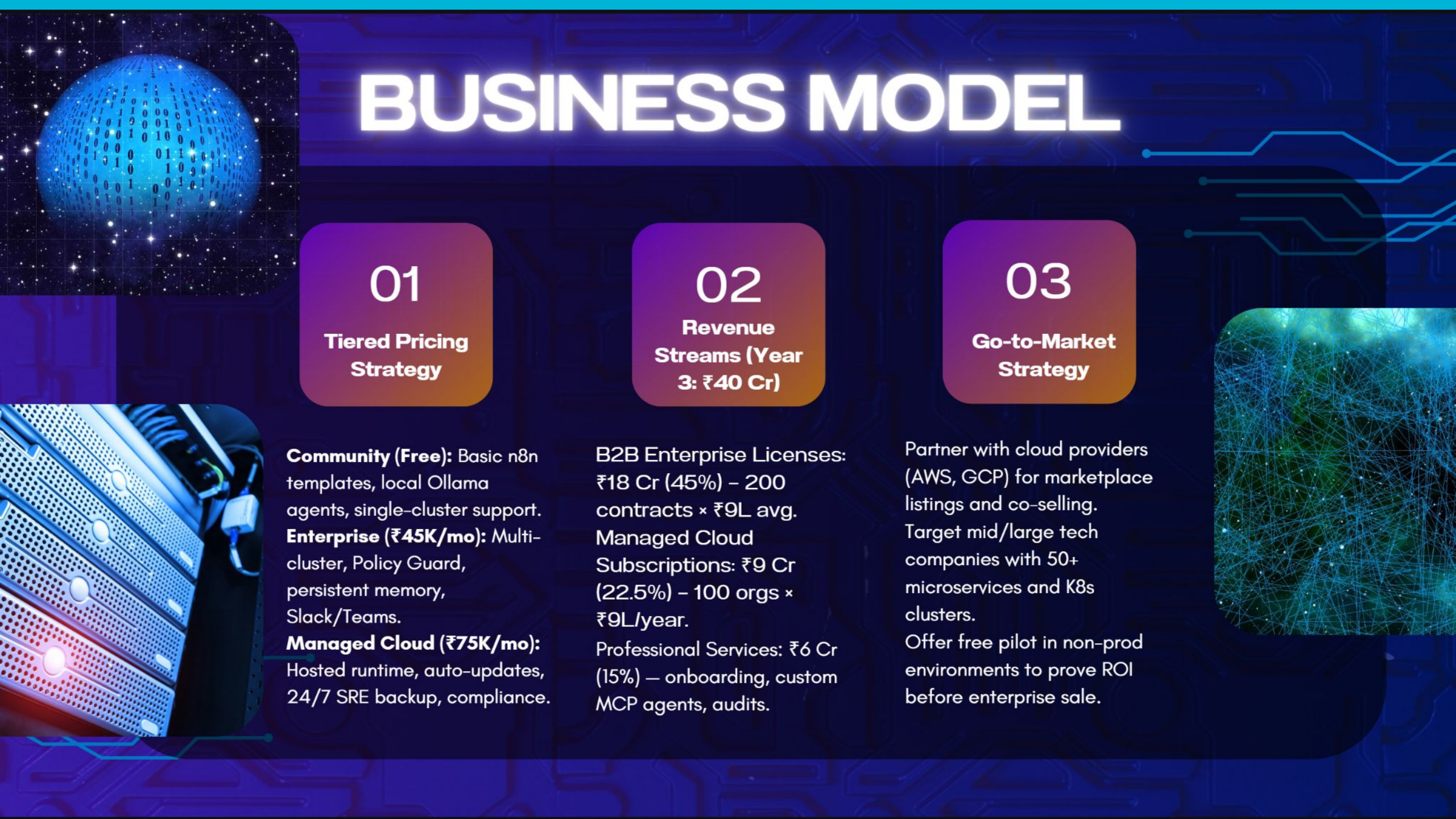


WHO BENEFITS



CTO & Engineering Leadership

MTTR drops 65%, engineering velocity increases, downtime costs plummet. \$300K+/hour outage prevention delivers immediate ROI on implementation. Teams freed from firefighting to focus on innovation and strategic initiatives.



BUSINESS MODEL

01

Tiered Pricing Strategy



Community (Free): Basic n8n templates, local Ollama agents, single-cluster support.

Enterprise (₹45K/mo): Multi-cluster, Policy Guard, persistent memory, Slack/Teams.

Managed Cloud (₹75K/mo): Hosted runtime, auto-updates, 24/7 SRE backup, compliance.

02

Revenue Streams (Year 3: ₹40 Cr)

B2B Enterprise Licenses: ₹18 Cr (45%) – 200 contracts × ₹9L avg.

Managed Cloud Subscriptions: ₹9 Cr (22.5%) – 100 orgs × ₹9L/year.

Professional Services: ₹6 Cr (15%) – onboarding, custom MCP agents, audits.

03

Go-to-Market Strategy



Partner with cloud providers (AWS, GCP) for marketplace listings and co-selling.

Target mid/large tech companies with 50+ microservices and K8s clusters.

Offer free pilot in non-prod environments to prove ROI before enterprise sale.

INNOVATION & MOAT



Agentic Self-Healing (Not Just Alerting)

AI doesn't just detect and alert — it diagnoses, patches, and redeploys autonomously.

OpenClaw agents bind LLM reasoning to K8s/n8n APIs for real action, not just insights.

Barrier: 3-4 years ahead of traditional AIOps monitoring tools.



LOCAL-FIRST, AIR-GAPPED DIAGNOSIS

Runs entirely on-prem/VPC — no sensitive logs leave enterprise environment.

OpenClaw local runtime hosts Ollama/Whisper models for privacy-compliant AI.

Barrier: 2-3 years — cloud SaaS competitors can't match air-gapped capability.



Persistent Memory & Continuous Learning

Learns from every outage, avoids repeating bad patches across clusters.

OpenClaw memory feeds continuous fine-tuning and heuristic updates automatically.

Barrier: 4-5 years — network effects strengthen with each deployment.



SCALABILITY & GROWTH



1. Domain Expansion Roadmap

Phase 1: CI/CD + Kubernetes self-healing (current).

Phase 2: Data Pipelines (Airflow/dbt), Infrastructure (Terraform/Ansible).

Phase 3: Application-level self-healing (auto-scaling, circuit breakers, retries).



2. Systemic Integration Strategy

Bundle with GitOps tools (ArgoCD, Flux), Service Mesh (Istio, Linkerd).

Integrate with observability platforms (Grafana/Loki, Datadog, New Relic).

OpenClaw agents auto-generate post-incident reports and compliance tickets.



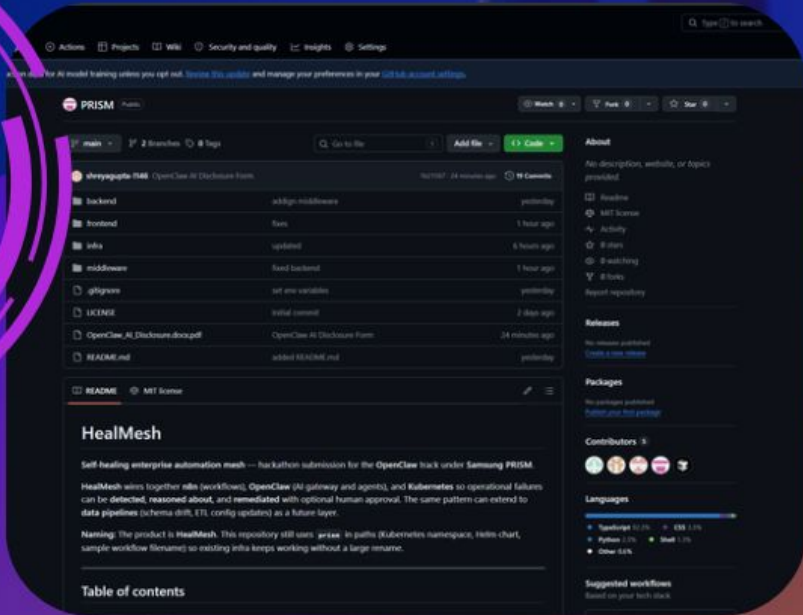
3. Tech Leverage for Scale

eBPF for low-overhead log streaming without performance degradation.

GitOps for declarative, auditable patching across thousands of clusters.

OpenClaw local runtime scales from single-node dev to multi-region K8s fleet.

DEMO LINK



GITHUB REPOSITORY

<https://github.com/Anweshal1111/PRISM/tree/main>



DEMO

[linhttps://drive.google.com/file/d/1jVepwzBg9AojgRfx29Umf56BWVOK3I95/view?usp=sharing](https://drive.google.com/file/d/1jVepwzBg9AojgRfx29Umf56BWVOK3I95/view?usp=sharing)

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: HealMesh

Project / Product Name: SELF-HEALING ENTERPRISE AUTOMATION MESH

Organization / Institution (if any): R. V. College of Engineering

Submission Date: 8th May 2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? Yes / No

Yes

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: Architecture mapping, agent workflow design, failure pattern research

Code generation or assistance: nifi JSON scaffolding, Python middleware stubs, Helm/K8s YAML templates, shell script logic

UI / UX design: Minimal (dashboard wireframes only)

Content creation: CONTRACTS.md, HANDOFF.md, README structure, policy documentation

Data analysis: Log parsing, incident payload validation, K8s event correlation

Testing / debugging: Webhook smoke tests, Helm deployment troubleshooting, nifi execution tracing

Other: Prompt engineering, agent behavior tuning, RBAC scope validation

DOCUMENT

AI DISCLOSURE FORM



WHY BET ON US

01

**Massive
Market
Opportunity**



\$50B+ DevOps & Data Ops market — largest untapped automation frontier. 68% of engineering hours wasted on firefighting represents huge ROI potential. First movers in offline-first, multi-stack, agentic self-healing for enterprises.

02

**Technical
Differentiation**

OpenClaw solves air-gapped diagnosis problem no cloud SaaS can touch. Persistent memory + policy-aware agents = 3× faster MTTR vs traditional AIOps. ₹40 Cr Year-3 revenue with ₹0 egress cost for the AI diagnosis layer.

03

**Privacy &
Compliance
Ready**



Local-first architecture ensures logs and configs never leave VPC. SOC2/GDPR compliant out of the box, DPDP Act ready for Indian enterprises. Open-source, auditable codebase builds trust with security-conscious CIOs.

THANK YOU!

Build the future of autonomous infrastructure – one self-healing agent at a time.

Anwasha Mohapatra

Shreya Gupta

Kasvi Gupta

Elina Somnath